

Programming the Pico Computer 3 in C

Fuzix on the RP2350B

What this is

Everything a C programmer needs to reach the facilities this port adds to Fuzix: the graphics modes, the off-screen framebuffer, text at pixel positions, sound — both the BBC synthesiser and streamed PCM — the shared maths library, the joystick and ADC, and the 8 MiB of PSRAM.

None of it is a library call. The kernel owns the hardware and exposes it through **ioctls on /dev/sys**, so everything below is the same three lines:

```
#include <fcntl.h>
#include <sys/ioctl.h>
#include "pico_ioctl.h"           /* the numbers and structures */

int sys = open("/dev/sys", O_RDWR);
ioctl(sys, GFXIOC_MODE, &mode);
```

`pico_ioctl.h` is the authority. If this document and that header disagree, the header is right.

Two ways to write C here

There are two, and they are genuinely different environments.

On the board: cc

```
$ cc -o prog.bc prog.c
$ ./prog.bc
```

The Fuzix C compiler runs on the machine itself and produces bytecode with native ARM Thumb spans, executed by `bcrun`. This is what `mmbc` targets and what most PC3 programs are.

What you get:

- headers: `stdio.h`, `stdlib.h`, `string.h`, `assert.h`, `limits.h`, `stddef.h`, `math.h`, `ctype.h`, `stdint.h`, `time.h` — in `/usr/lib/cc/include`, describing what `bcrun` provides rather than what Fuzix's own `libc` does.

- a C library subset resolved **by name at load time**: `printf`, `sprintf`, `fprintf`, `fopen`/`fclose`/`fread`/`fwrite`/`fgetc`/`fgets`/`fputc`/`fputs`/`fseek`/`ftell`/`feof`, `malloc`/`calloc`/`realloc`/`free`, the `str*` and `mem*` family, `atoi`/`atof`/`atol`/`strtol`/`strtod`, `open`/`close`/`read`/`write`/`lseek`/`creat`/`remove`/`rename`/`unlink`, `exit`, `rand`/`srand`, `time`. From v0.10 the `str*` and `mem*` family are **newlib**'s, taken from the ARM toolchain's own `libc.a` — tuned assembly rather than the byte-at-a-time loops Fuzix ships generically. A `memcpy` that used to run at about nine cycles a byte now moves a word at a time. Nothing in a program needs changing to get it, but a program whose profile was dominated by string work will be noticeably quicker.
- two of our own: `adval(n)` and `time_us()` / `time_us64()`.
- the MMBasic runtime, callable from plain C: `mmb_runtime.h` declares the `mm_*` entry points `bcrun` resolves by name — `PRINT` and formatting, strings, files, and the graphics crossings — and the drawing primitives are static functions in one header per primitive: `mmb_gfx_circle.h`, `mmb_gfx_box.h`, `mmb_gfx_rbox.h`, `mmb_gfx_triangle.h`, `mmb_gfx_arc.h`, `mmb_gfx_text.h`, `mmb_gfx_map.h` (batch helpers in `mmb_gfx_pts.h`; `mmb_gfx.h` is the umbrella that includes the lot; `mmb_gpio.h` is `SETPIN/PIN`). Include only what you draw: the compiler drops an unused static, but the rule counts names rather than reachability, so a recursive primitive survives inside any header that carries it — the include is the granularity.

Because names resolve at load time, **declaring a function is all the header you need**:

```
extern long time_us(void);           /* no header required */
```

The catch: **ioctl is not in that table**. A `bcrun` program cannot call it, so it cannot reach the graphics or PSRAM ioctls directly. What it gets instead is `adval()` and `time_us()`, which `bcrun` implements on its behalf. To drive the display from a `bcrun` program, either write MMBasic and let `mmbc` translate it, or call the `mm_*` runtime and the `mmb_gfx_*.h` primitives from C directly (the bullet above) — either way the runtime does the ioctls.

Cross-compiled: native ARM binaries

Everything in `Kernel/platform/platform-rpipico/Utils/` is built this way, and this is the path for a program that needs the full syscall interface:

```
arm-none-eabi-gcc -mcpu=cortex-m33 -Os -fno-builtin \
    -isystem $FUZIX/Library/include -c prog.c
arm-none-eabi-ld $FUZIX/Library/libs/crt0_armm0.o prog.o -o prog \
    -L$FUZIX/Library/libs -lcarmm0 -pie -static -no-dynamic-linker \
    -z max-page-size=4 -T $FUZIX/Library/elfexe32.ld
```

or just add it to `Utils/Makefile` and `make`. These get Fuzix's own `libc`, `ioctl`, and every syscall. **Everything in the rest of this document assumes this**

path.

Ask for more stack with `-z stack-size=N` if you recurse; the default is 8 KB and the ELF loader honours the request up to 64 KB.

Floating point is soft float, and that is not negotiable for most programs. The RP2350's FPU is granted, but no FP register state is saved across a context switch, so exactly one process on the machine may execute FP instructions. That process is the audio player, and it holds the sound device for as long as it runs — the device lock *is* the FPU lock. Do not add `-mfpu` to anything else. For maths that has to be fast, use the shared library below: it is double precision, it runs on the DCP, and it costs no lock.

Worked examples on disc

Read these before writing anything; they are all short.

program	shows
utils/gfxtest.c	modes, palette, GFXIOC_BLIT
utils/linebench.c	GFXIOC_PIXEL against GFXIOC_PIXELS, with timings
utils/ripple.c	per-pixel drawing versus a shadow buffer
utils/bmtest.c	GFXIOC_BITMAP and GFXIOC_RECT, verified by readback
utils/saveimage.c, loadimage.c	reading and writing the screen
utils/memprobe.c	how much memory a process can actually have
utils/allocbench.c	the cost of a PSRAM arena allocation
utils/pcmtest.c	the PCM sound sink, from a generated tone
utils/playmp3.c	a complete decoder: file → PCM → the ring
utils/libmbench.c	the shared maths library against the local one

The memory a program gets

- The process pool is **340 KB**, and one process may use nearly all of it — `memprobe` reports about **292 KB**. The ceiling is not a constant: a grow must leave room for the largest *other* process to be resident (`largest_neighbour()` in `swapper.c`), so it moves with what else is running.
- Up to **64 processes**.

- Memory is packed in 4 KB chunks at actual size. A swapped-out process gets a PSRAM allocation of exactly its own size.
- `sbrk()` grows the break and fails when there is no room. Probing with `sbrk(4096)` in a loop until it refuses is a legitimate way to find your ceiling — `bbcbasic` does exactly that.
- **There is no MMU.** A wild pointer corrupts the kernel and takes the machine down with no diagnostic. This is the single most important thing to know.

Graphics

Modes

```
int mode = 7;
ioctl(sys, GFXIOC_MODE, &mode);      /* pass a POINTER to an int */
```

mode	resolution	colours	raster
0, 3	640×256, 1bpp	2	1024×768
1, 4	320×256, 4bpp	16	1024×768
2, 5	160×256, 4bpp	16	1024×768
7	320×240, 4bpp	16	640×480
0xFF	back to the 80×40 text console		640×480

Modes 0–5 are the BBC Micro set. **Mode 7 is not teletext** — it is 320×240 in 16 colours, and it is what MMBasic calls `MODE 2`.

Switching *within* a raster keeps the monitor locked; crossing between 640×480 and 1024×768 restarts the scanout and the monitor resyncs.

Geometry, so nothing has to hardcode it:

```
struct gfx_info gi;
ioctl(sys, GFXIOC_INFO, &gi);
/* gi.width, gi.height, gi.stride, gi.bpp, gi.mode */
```

Colour

Everything speaks **RGB888** and the kernel converts to the mode’s own colours — you never deal in colour indices.

```
ioctl(sys, GFXIOC_COLOUR, (void *)0xFF8000L); /* the VALUE, not a pointer */
```

Mode 7’s sixteen colours are MMBasic’s RGB121 set: red and blue are 0 or 255, green has four levels. Anything else maps to the nearest, which is worth knowing before you ask for `RGB(r, g, 99)` and get black — a blue of 99 is nearer 0 than 255 every time.

Drawing

```
ioctl(sys, GFXIOC_PIXEL, (void *)GFX_PIXEL_PACK(x, y)); /* current colour */
int c = ioctl(sys, GFXIOC_GETPIXEL, (void *)GFX_PIXEL_PACK(x, y));

struct gfx_rect r = { x1, y1, x2, y2 };
ioctl(sys, GFXIOC_RECT, &r); /* filled, current colour */
```

GFX_PIXEL_PACK puts the coordinates in the *ioctl* argument, so there is nothing to copy in and nothing to validate: measured at 1.30 μ s against 1.488 μ s for an ordinary *ioctl*. Its *y* is 9 bits, so it stops at 511 — use the batch calls for the 768-line modes.

A syscall costs 1.3 μ s and a pixel store costs 15 ns. Anything with more than a handful of points should be batched:

```
struct gfx_pt pts[512];
struct gfx_batch b = { .count = n, .flags = 0,
                      .items = pts, .colours = NULL };
ioctl(sys, GFXIOC_PIXELS, &b); /* a run of points */
ioctl(sys, GFXIOC_RECTS, &b); /* a run of rectangles: gfx_rc[] */
```

colours may be NULL for “all in the current colour”, or one RGB888 per item. Up to GFX_BATCH_MAX (1024) items. A 312-point line costs 433 μ s one at a time and 71 μ s batched.

The arrays are read where they lie — nothing is copied into the kernel.

Bitmaps and text

```
struct gfx_bitmap gb = { x, y, w, h, scale, 0, fg, bg, bits };
ioctl(sys, GFXIOC_BITMAP, &gb);
```

Bits are MSB-first, row-major — **exactly** MMBasic’s font packing, so fonts interchange. *bg* = -1 is transparent. Source up to GFX_BITMAP_MAX.

For text there is no need to carry a font — there are nine of them in the kernel, MMBasic’s own, and they are *const* so they live in flash and cost a program nothing:

```
struct gfx_text gt = { x, y, scale, font, fg, bg, len, str };
int endx = ioctl(sys, GFXIOC_TEXT, &gt);
```

This draws a run of characters and returns the *x* it ended at. *bg* = -1 leaves the paper alone. Up to GFX_TEXT_MAX (256) characters, one crossing. *font* is 1..9 as MMBasic numbers them, and **0 means 1**, so code written before there were fonts to choose from still gets the console’s:

font	cell	characters
1	8×12	224 — the console’s own

font	cell	characters
2	12×20	95
3	16×24	95
4	10×16	224
5	24×32	95
6	32×50	11 — the digits only
7	6×8	96
8	4×6	96
9	8×10	95

A character the font does not have fills its cell with the paper colour, which is what MMBasic does. That is not a corner case: with font 6 it is every character that is not a digit.

Ask for the cell rather than assuming it — laying text out against the wrong metrics is the whole failure mode here:

```
struct gfx_fontinfo fi = { .font = 3 };
ioctl(sys, GFXIOC_FONTINFO, &fi);
/* fi.width, fi.height, fi.first, fi.count, fi.nfonts */
```

fi.width comes back 0 for a font that does not exist.

The palette

A 4bpp mode stores a colour *number* per pixel and the palette says what each number is, so remapping recolours everything already drawn without touching a byte of the framebuffer. This is MMBasic's MAP.

```
ioctl(sys, GFXIOC_MAP, (void *)((index << 24) | rgb888)); /* collect */
ioctl(sys, GFXIOC_MAPCTL, (void *)GFX_MAP_SET); /* apply */
ioctl(sys, GFXIOC_MAPCTL, (void *)GFX_MAP_RESET); /* default */
```

GFXIOC_MAP changes nothing on screen: entries are collected and GFX_MAP_SET moves the whole palette across **during the vertical blanking**, so a new palette arrives between frames rather than recolouring the picture in instalments. 16-colour modes only.

The colour stored is RGB332 — the byte the scanout puts on the wire — reduced by truncation, as MMBasic's RGB332() does.

Note that this does not change which entry a colour lands on. In the 320x240 mode that is fixed bit extraction (MMBasic's RGB121: one bit of red, two of green, one of blue), deliberately independent of the palette, so a program can remap freely and go on naming colours the same way.

```
ioctl(sys, GFXIOC_SCROLL, (void *)((rows << 24) | rgb888));
```

Scrolls the write target: **rows** signed in the top byte, positive up. This is the same call the console uses for its own scrolling, so a program and the shell move the picture the same way.

The off-screen framebuffer

Draw off-screen, show it in one go. This is MMBasic's FRAMEBUFFER.

```
ioctl(sys, GFXIOC_FBOPEN, (void *)1);    /* claim it */
ioctl(sys, GFXIOC_FBSEL, (void *)1);     /* draw off-screen */
... every drawing call above now goes to the buffer ...
ioctl(sys, GFXIOC_VSYNC, (void *)0);     /* optional: top of frame */
ioctl(sys, GFXIOC_FBCOPY, (void *)0);    /* buffer -> screen */
ioctl(sys, GFXIOC_FBSEL, (void *)0);     /* back to the screen */
ioctl(sys, GFXIOC_FBOPEN, (void *)0);    /* give it back */
```

Things to know:

- **The layer is owned.** FBOPEN fails with EBUSY if another process holds it, EINVAL on a board with no PSRAM. It is released automatically on exit, on exec, and on a mode change — so a program that dies does not leave the machine drawing into nowhere.
- **The write target is per process.** Anything else that draws while you are between frames — another program, the shell — still goes to the screen. You cannot have your picture scribbled on.
- **CREATE belongs after MODE.** A mode change discards the buffer, because its contents are in the geometry of the mode being left.
- FBCOPY takes 1 to copy screen→buffer instead.

What it costs, measured on the board in mode 7:

full-screen fill, to the screen (SRAM)	1.49 ms
the same into the buffer (PSRAM)	3.63 ms
copy the 38,400-byte buffer to the screen	0.73 ms (53 MB/s)

So a redraw-and-show frame is about 4.4 ms against the 17 ms the display gives you. GFXIOC_VSYNC before the copy holds a loop to the refresh rate — 59 fps with room for roughly three times as much drawing.

Sound

Four channels, BBC SOUND semantics.

```
struct snd_cmd s = { .chan = 1, .amp = -15, .pitch = 100, .dur = 20 };
ioctl(sys, SNDIOC_SOUND, &s);    /* -1/EAGAIN if that queue is full */
ioctl(sys, SNDIOC_ENV, envbytes); /* 14 bytes: N,T,PI1..3,PN1..3,AA,AD,AS,AR,ALA,ALD */
ioctl(sys, SNDIOC_QUIET, 0);     /* silence everything */
```

amp is -15..0 for volume, or 1..16 to use ENVELOPE n. The channel word carries the &1x flush and &Sxx sync bits.

Playing your own samples

The same I2S engine will take decoded PCM instead of the synthesiser. Open the stream, write 16-bit frames as fast as the ring will take them, and the driver's DMA does the rest — nothing has to be refilled from a main loop, which is what lets music carry on while another program runs.

```
struct snd_pcm cfg = { .rate = 44100, .channels = 2, .bits = 16 };
struct snd_buf sb;
struct snd_stat st;

if (ioctl(sys, SNDIOC_PCMOPEN, &cfg) < 0)    /* EBUSY: someone else has it */
    ...
sb.base = samples; sb.len = nbytes;
n = ioctl(sys, SNDIOC_PCMWRITE, &sb);        /* returns bytes ACCEPTED */
ioctl(sys, SNDIOC_PCMSTAT, &st);             /* space, queued, underruns */
ioctl(sys, SNDIOC_PCMLCLOSE, 0);
```

- A **short write is normal**, not an error: it means the ring is full and you should come back. The ring holds 256 KB, about 1.5 seconds at 44100 stereo — deep on purpose, because a Fuzix process can be swapped out for tens of milliseconds and the audio must not notice.
- **Mono is expanded by the driver**, so a mono source costs you nothing and halves the traffic into the ring.
- **underruns** counts half-buffers the interrupt had to fill with silence. It is the objective test of whether your buffering is deep enough, and much better than “it sounded all right”.
- To play the tail out, poll PCMSTAT until **queued** is zero and close then; PCMLCLOSE itself is immediate and drops whatever is left.

One process at a time. There is one I2S engine, so the stream belongs to whoever opened it: a second PCMOPEN gets EBUSY, and writes and closes from anyone else are refused. This is deliberate — two players sharing the ring interleave their samples, and it sounds precisely as bad as that suggests.

```
int who = ioctl(sys, SNDIOC_PCMOWNER, 0);    /* the pid playing, or 0 */
```

SNDIOC_PCMOWNER is how a program that did not start the player can find it: send it SIGINT to stop the music tidily. A player that dies without closing strands nothing — the kernel hands the stream back when its owner is gone, so SIGKILL is safe too.

SOUND and PCM are mutually exclusive, as they are in MMBasic: opening the stream silences the synthesiser and closing it hands the state machine back.

The shared maths library

`sin`, `cos`, `exp` and the rest live in kernel flash and are called **directly** — there is no MMU, so kernel flash is in the same address space as your program. Ask for the table once:

```
void *tbl;  
ioctl(sys, PICOIOC_LIBM, &tbl);      /* check magic and version first */
```

It is double precision and uses the RP2350's DCP, which makes it about **2.7×** **faster** than the soft-float library linked into a program, and it costs nothing in program memory. `libm_table.c` documents the layout and the errno contract (these do not report domain errors).

Check the magic and version before calling anything. An old binary on a new kernel must fail rather than call the wrong slot — that is what the version is for.

Pins, the joystick and the ADC

These are **not** syscalls. Claim what you want through the pin lock, then read and write the registers yourself — there is no MMU on this board, so a pin costs a store rather than the 1.488 μ s an `ioctl` costs. `<sys/pc3io.h>` has the registers and the claim wrappers:

```
#include <sys/pc3io.h>  
  
pc3_claim(PLK_PIN, 4);          /* one syscall, once */  
pc3_pin_out(4);  
pc3_pin_high(4);               /* one store */  
  
pc3_claim(PLK_PIN, 34);  
pc3_pin_in(34, 1);             /* pulled up */  
if (!pc3_pin_get(34)) ...      /* the joystick's bit 0, active low */  
  
pc3_claim(PLK_ADC, 0);  
pc3_claim(PLK_PIN, PC3_ADC_GPIO(1));  
pc3_adc_enable();  
pc3_adc_pin(1);  
v = pc3_adc_read(1) << 4;     /* ADVAL's 16-bit convention */
```

Claimable: header pins GP0–GP7, GP26, GP34–GP46; I2C1; SPI0; the twelve PWM slices; the ADC. Everything else belongs to the board. A claim you already hold succeeds; someone else's gives `EBUSY`; anything not on that list gives `EINVAL`. **Exiting releases everything and resets the pins**, so a program that dies driving a relay does not leave it driven — which is the reason the kernel is involved at all.

Two traps, both of which produce code that looks right and does nothing:

- **PADS_BANK0.ISO resets to 1.** Until it is cleared the pad is disconnected — no drive, no read, and a pull-up does nothing. Every setup function in the header clears it last. If you write your own, do the same.
- **Never read-modify-write SIO's GPIO_OUT or GPIO_OE.** They are shared with the kernel and with core1's display. Use the SET/CLR/XOR registers, and remember GP32–GP47 are a second bank whose registers interleave with the first.

`utils/locktest.c` exercises the whole thing, including killing a process that holds a pin and checking the pin comes back.

ADVAL, and the clock

```
int v = ioctl(sys, PICOIOC_ADVAL, (void *)n);
```

n	reads
0, 1–4	gone — joystick and ADC are pin work, see above
-5..-8	free slots in sound queue 0–3
-9	microsecond counter, 31 bits in the return value
-10	the full 64-bit counter: pass an 8-byte buffer whose low word holds -10

BASIC's `ADVAL(0)` and `ADVAL(1)–ADVAL(4)` are unchanged: BBC BASIC reads the pins itself now, and returns exactly what it used to.

PSRAM

8 MiB, outside the process image, reached as a raw address.

```
struct psram_req rq = { .len = 64 * 1024 };
ioctl(sys, PSRAMIOC_ALLOC, &rq);      /* rq.base is the address */
ioctl(sys, PSRAMIOC_REALLOC, &rq);    /* rq.base in, the NEW base out */
ioctl(sys, PSRAMIOC_FREE, (void *)rq.base);
```

```
struct psram_stat st;
ioctl(sys, PSRAMIOC_STAT, &st);        /* total, free, largest */
```

- Allocation is 4 KB granular and **released on exec and exit**; a fork leaves the arena with the parent.
- **REALLOC may move the block** — newlib copies when it cannot extend in place — so rebuild any interior pointers.
- The address is raw and unprotected; there is no MMU.

- An alloc+free pair costs about **5.1 μ s**, against 350 ns for an in-process malloc. Ask once and sub-allocate, rather than per call.
- **PSRAM is roughly 2.4 $\times$ slower than SRAM** for scattered access, but a bulk sequential copy runs at 53 MB/s. Put big, streamed things there; keep hot, randomly-accessed things in your own memory.

Miscellaneous

```
ioctl(sys, PICOIOC_FLASH, 0);           /* reboot into BOOTSEL */
ioctl(sys, PICOIOC_KBDMAP, "uk");       /* US UK DE FR ES BE */

int board;
ioctl(sys, PICOIOC_BOARD, &board);     /* 2 or 3 */
```

PICOIOC_BOARD answers which machine this is — the same number the boot banner prints, from the same detection (the DS3231's 32 kHz on GP27). It exists because there was no other way to ask: the kernel knew, and only the banner ever said. A program that wants to name itself needs it — MMBasic's MM.DEVICE\$ is the first caller, and reports **Fuzix on PC2** or **Fuzix on PC3**.

An older kernel does not have it, so check the return value and pick a default rather than trusting whatever was in the variable.

Traps worth knowing

No MMU. Said once already, worth saying twice. A bad pointer in a user program takes the whole machine down with no message.

A panic blanks the HDMI screen. plt_monitor() resets core1 and with it the scanout, so a panic looks like a dead display. **The message only ever goes out of the serial port.** If a program kills the machine, read the UART.

Text and graphics share the framebuffer. In a graphics mode the console draws into the screen buffer, so printf from your program lands on the picture and scrolls it when the line wraps. Use GFXIOC_TEXT for text you want positioned, and keep printf for the serial console.

Timing is contended. core1 DMAs scanlines out of SRAM continuously, so RAM bandwidth is shared. This cuts both ways: it is why the kernel now executes from flash at no measurable cost, and why a benchmark that touches the framebuffer will not be reproducible to the last percent.

The compiler is small. cc on the board has real limits — deeply nested expressions and very large functions can exceed it. It also has a threshold beyond which a function stops being compiled to native ARM and falls back to bytecode, which is worth 2–3 $\times$ on a hot loop. If a small change makes a program suddenly

slower, that is the first thing to suspect; BCRUN_BYTECODE=1 forces bytecode so you can compare.

The preprocessor is the board's own. Cross-compiled builds use `gcc -E`, so anything that works there is not evidence that it works here. C89 is what to expect: `v0.10` fixed?: inside `#if`, `#line`, `__LINE__` inside `#if`, and argument expansion before `##` pastes — none of which had ever been tested, because no host build reaches this code. Measured C conformance on the board is 160 of 175 tests.

Unbounded recursion takes the machine down. `bcrun` checks the stack on entry to a bytecode function and stops the program cleanly, but a function the compiler turned into native ARM has a three- instruction prologue with no check in it — and a small recursive function is exactly what gets translated. A missing terminating condition is a reset, not an error message.

Where things live

Kernel/platform/platform-rpipico/	
pico_ioctl.h	every ioctl, and the authority
display.c/.h	modes, primitives, the framebuffer
console.c	the terminal, and the font
sound.c	the synthesiser and the PCM sink
libm_table.c	the shared maths library and its ABI
swapper.c	the process pool
psram.c, arena.c	PSRAM and the arena allocator
utils/	the worked examples above
PC3-GFX-DESIGN.md	why the graphics interface is shaped as it is
PC3-FLASH-PLAN.md	what runs from flash and what may not
FS32-FORMAT.md	the on-disk filesystem format, and its authority
BUILDING-PC3.md	building the kernel and the card